Computer Networks

Lecture 19 Data Link Layer

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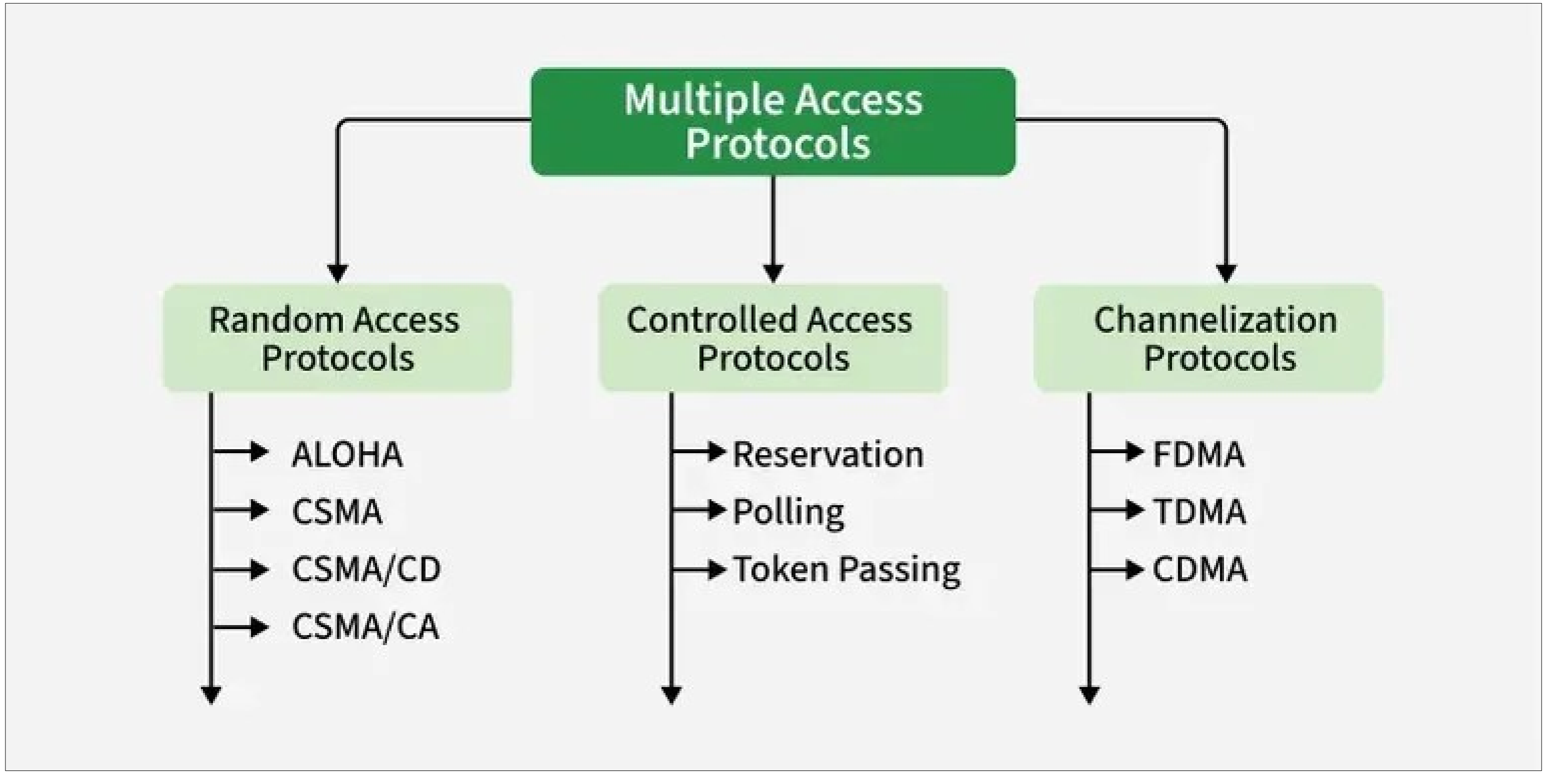
Content

- **Channelization Protocol**

Introduction

Many formal protocols have been devised to handle access to a shared link



Introduction

Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations.

Depending on how the bandwidth is divided, channelization can be implemented using different protocols.

Frequency Division Multiple Access (FDMA)

- Divides the channel into different frequency bands.

Time Division Multiple Access (TDMA)

- Divides the channel into time slots.

Code Division Multiple Access (CDMA)

- Assigns unique codes to different users to share the same channel simultaneously.

FDMA

Inn FDMA the total bandwidth of a communication channel is divided into several smaller frequency bands, and each user is assigned a separate frequency band for transmission.

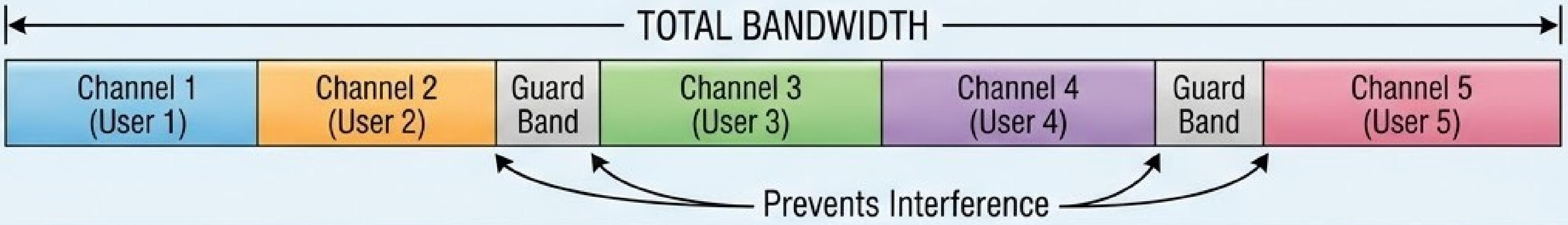
Example: Suppose a channel has 30 MHz bandwidth.

User	Frequency Band
User A	0 – 10 MHz
User B	10 – 20 MHz
User C	20 – 30 MHz

All three users transmit at the same time, but on different frequencies.

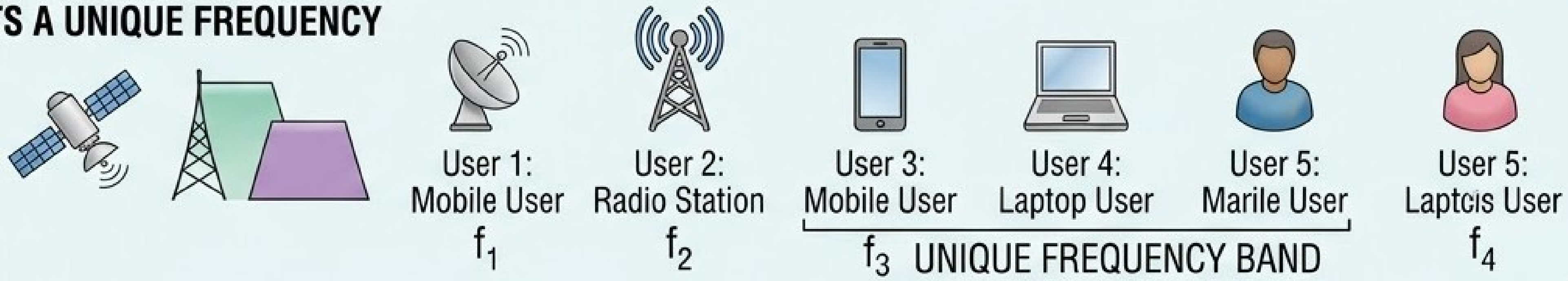
1. TOTAL AVAILABLE BANDWIDTH SPLIT INTO MULTIPLE FREQUENCY RANGES

The available channel bandwidth is split into multiple frequency ranges.



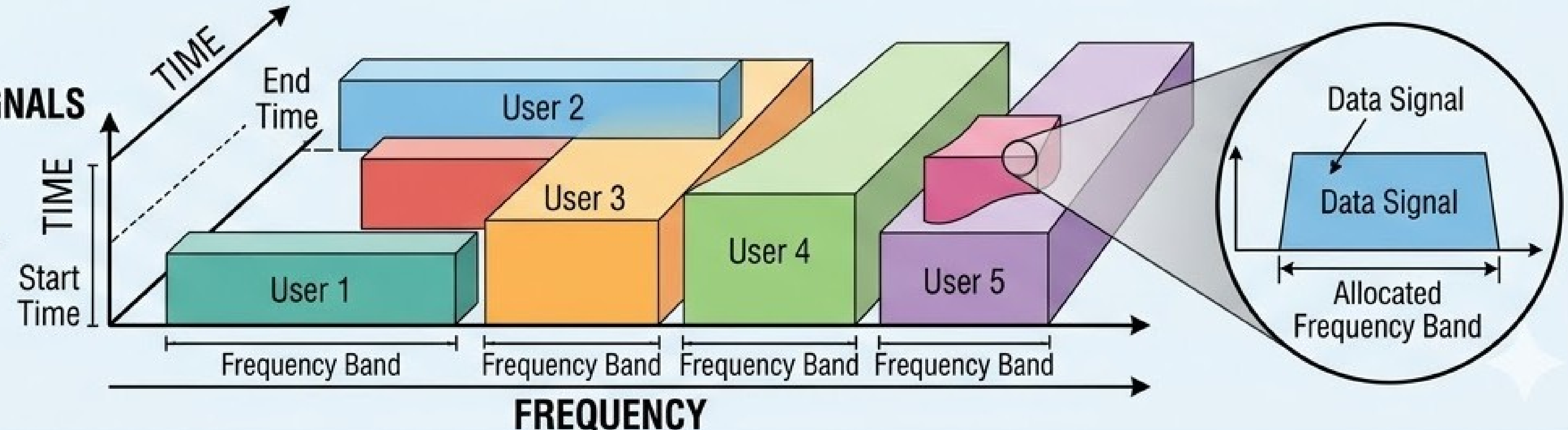
2. EACH STATION OR USER GETS A UNIQUE FREQUENCY BAND

Each station or user gets a unique frequency band.



3. ALL USERS TRANSMIT SIMULTANEOUSLY, BUT ON DIFFERENT FREQUENCIES, SO SIGNALS DO NOT INTERFERE

All users transmit simultaneously, but on different frequencies, so signals do not interfere.



TDMA

In Time-Division Multiple Access (TDMA) multiple users share the same communication channel by dividing time into slots.

Approach

- The entire channel bandwidth is available to all users.
- The time is divided into fixed time slots.
- Each station/user is assigned a specific time slot to transmit data.
- A user can transmit only during its assigned slot.
- When the slot ends, another user transmits.

Frame 1: | A | B | C | D |

Frame 2: | A | B | C | D |

Frame 3: | A | B | C | D |

CDMA

Code-Division Multiple Access (CDMA) is a channelization technique in which all users transmit over the same frequency band at the same time, but each user uses a unique code to encode their signal.

Steps

- All stations share the same bandwidth simultaneously.
- Each user is assigned a unique spreading code.
- Data is encoded using the assigned code before transmission.
- At the receiver, the same code is used to decode the intended signal and separate it from othe

CDMA

Idea

Assume there are four stations (1, 2, 3, and 4) connected to the same communication channel. Each station wants to transmit its own data. The data from station 1 is d_1, from station 2 is d_2, from station 3 is d_3, and from station 4 is d_4. Each station is also assigned a unique code: station 1 gets c_1, station 2 gets c_2, station 3 gets c_3, and station 4 gets c_4.

Important Properties of the Codes

The codes assigned to the stations have two important properties:

1. *Orthogonality:*

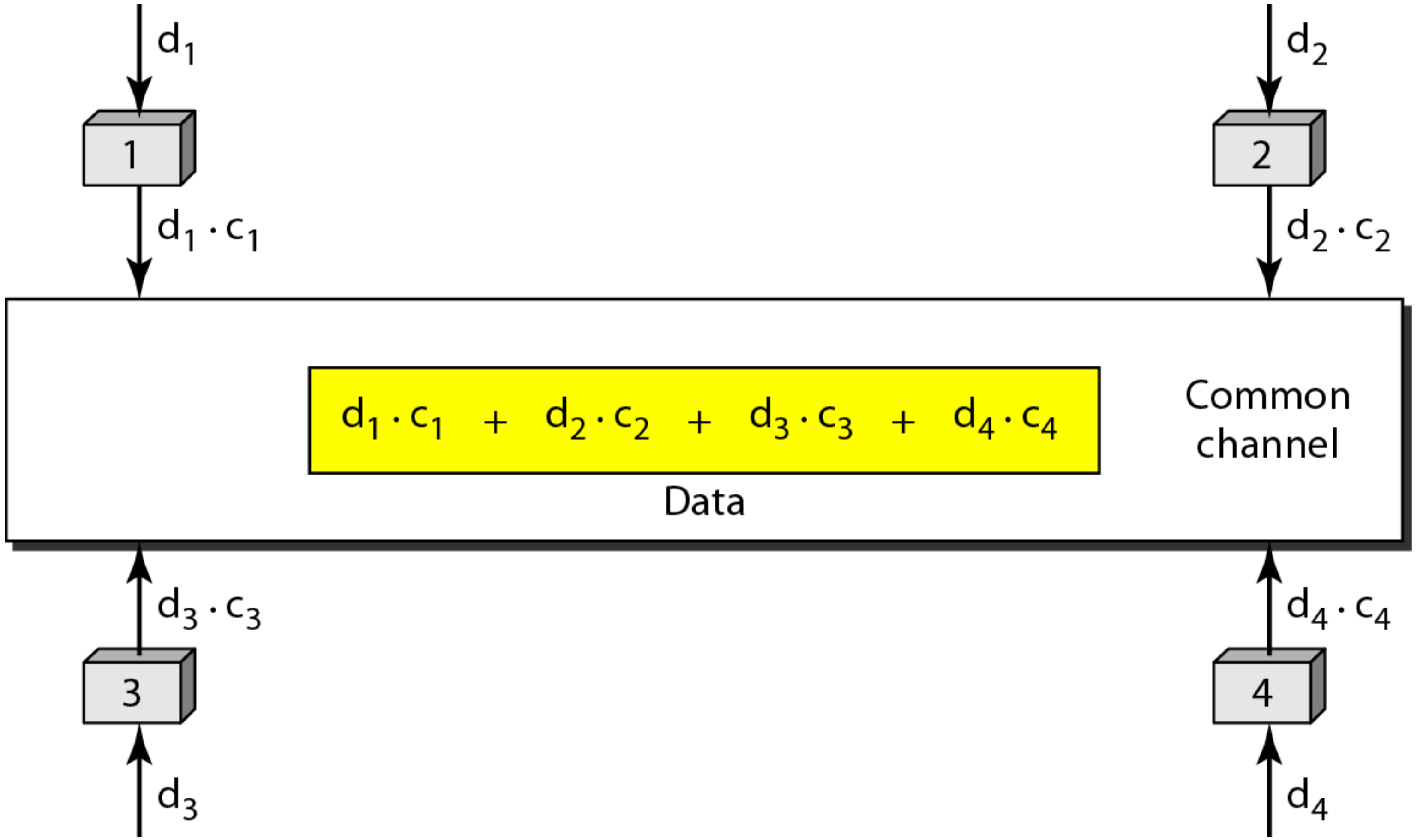
   If one station's code is multiplied by another station's code, the result is 0.

2. *Self-correlation:*

   If a station's code is multiplied by itself, the result equals 4 (which is the number of stations).

CDMA

Example



CDMA

Chips

In Code-Division Multiple Access (CDMA), each station is assigned a code, which is a sequence of numbers called chips.

- In CDMA, each user is assigned a unique code sequence.
- This code sequence is made up of smaller bits called chips.
- When a station sends a data bit (0 or 1), it multiplies the bit with the chip sequence.
- As a result, one data bit is converted into multiple smaller bits (chips) before transmission.

CDMA

Chips The chips are orthogonal sequences and have the following properties:

- Each sequence is made of N elements, where N is the number of stations.
- If we multiply a sequence by a number, every element in the sequence is multiplied by that element. This is called multiplication of a sequence by a scalar.

$$2.[+1\ +1-1-1]=[+2+2-2-2]$$

- If we multiply two equal sequences, element by element, and add the results, we get *N*, where *N* is the number of elements in the each sequence

$$[+1\ +1-1\ -1].[+1\ +1\ -1\ -1]\ =\ 1\ +\ 1\ +\ 1\ +\ 1\ =\ 4$$

- If we multiply two different sequences, element by element, and add the results, we get 0.

$$[+1\ +1-1\ -1].[+1\ +1+1+1]=\ 1\ +\ 1\ -\ 1\ -\ 1\ =0$$

CDMA

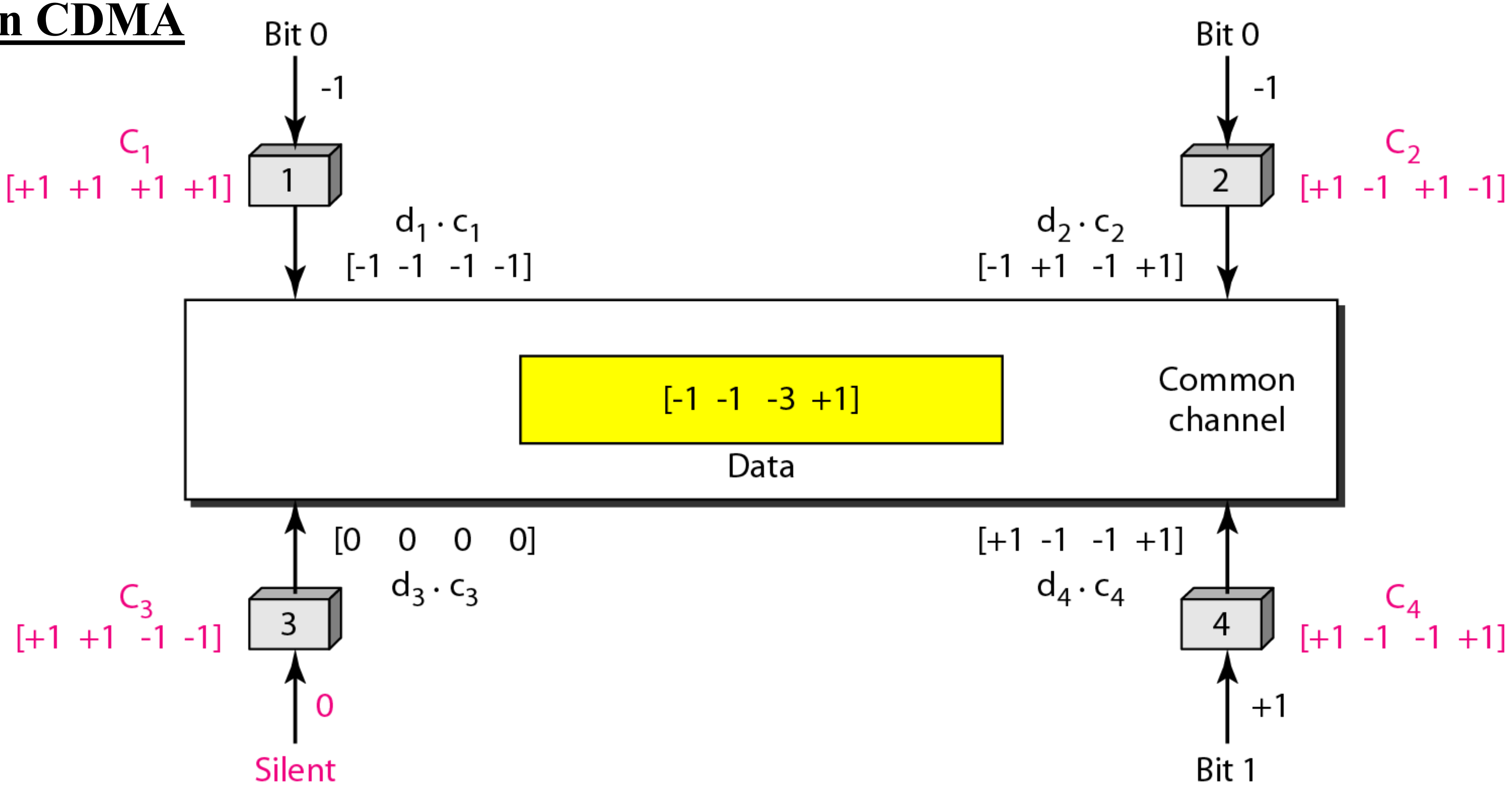
Data Representation

We follow these rules for encoding:

- If a station needs to send a 0 bit, it encodes it as -1;
- if it needs to send a 1 bit, it encodes it as +1.
- When a station is idle, it sends no signal, which is interpreted as a 0.

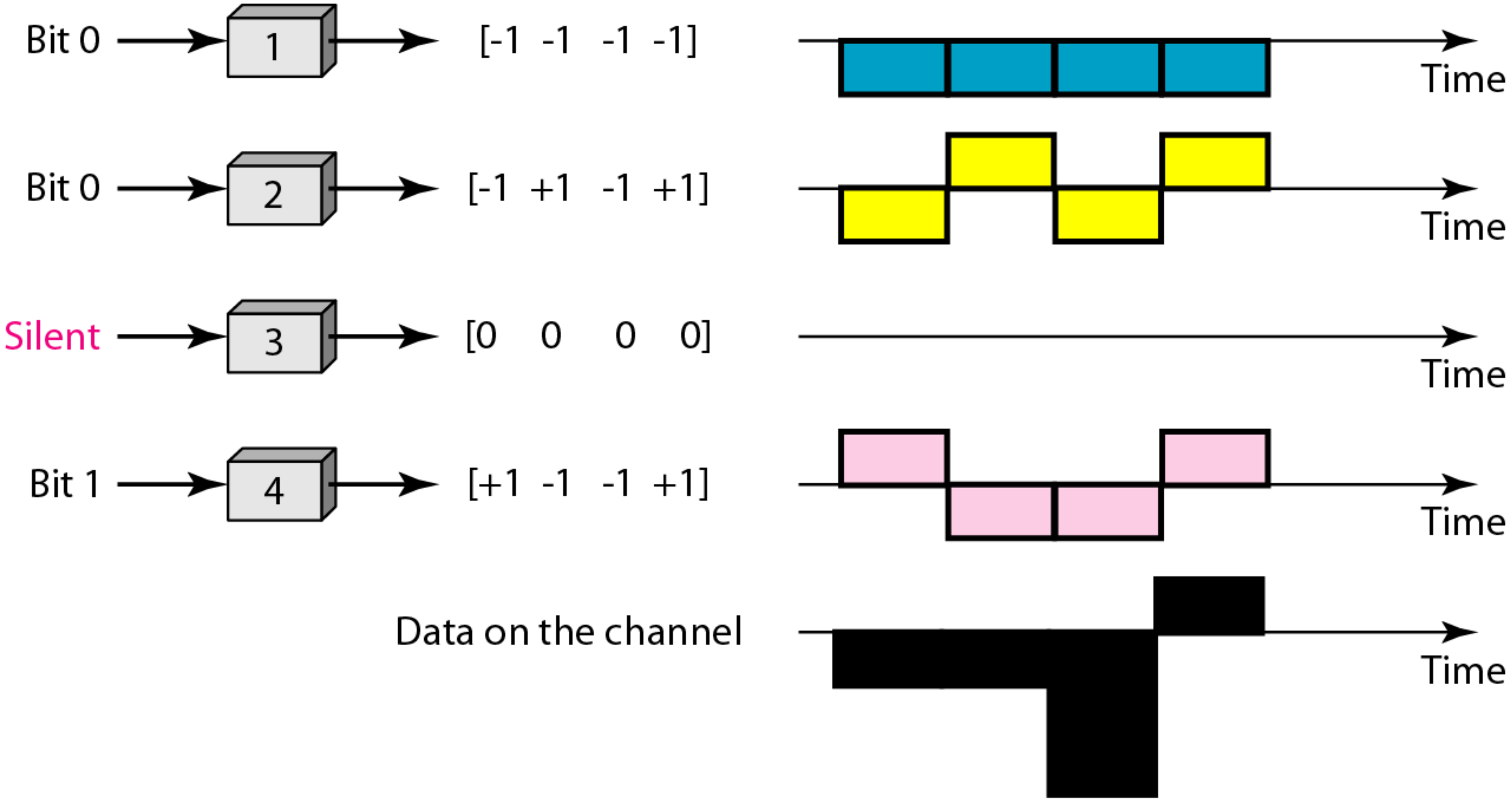
CDMA

Sharing channel in CDMA



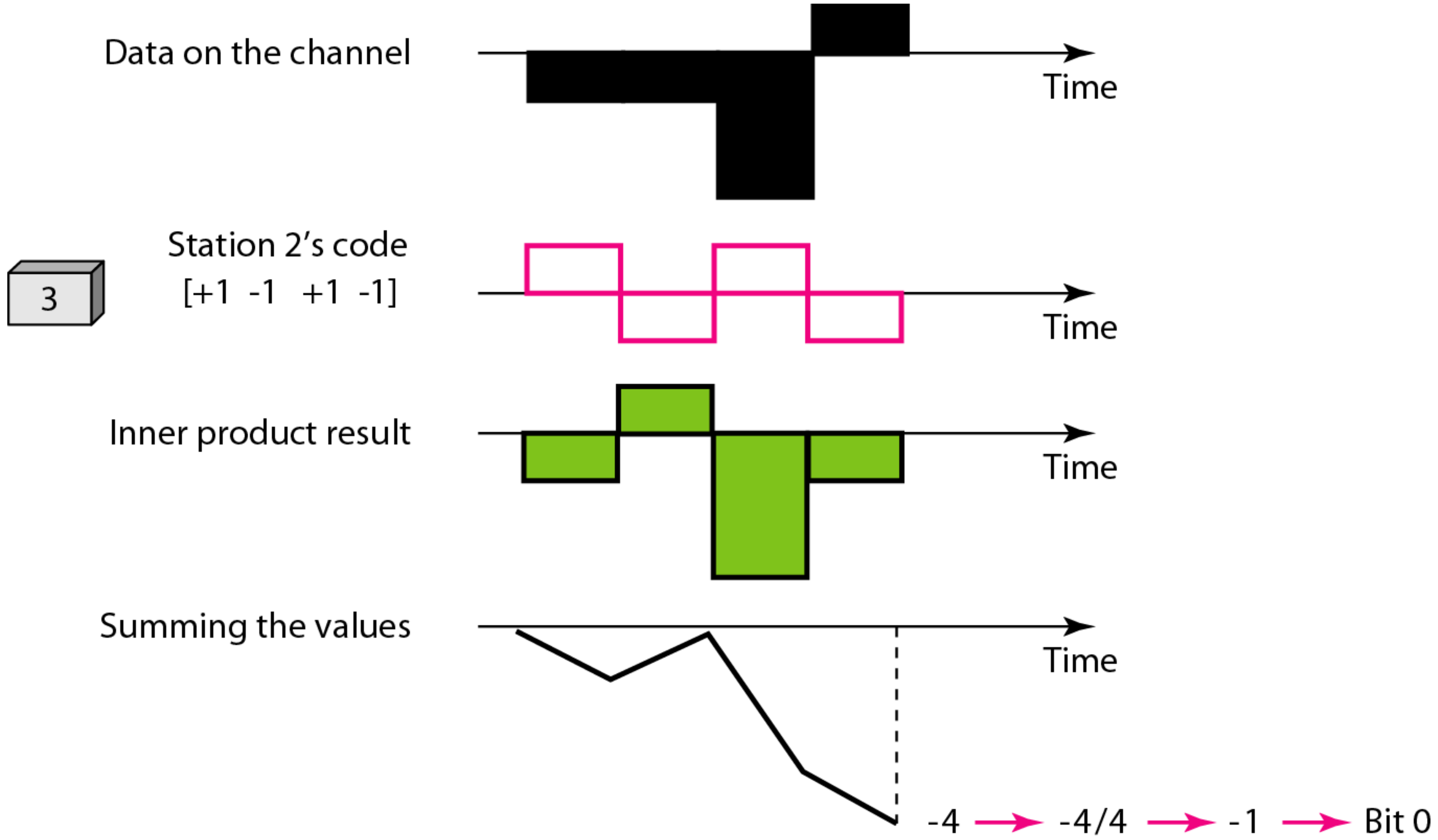
CDMA

Sharing channel in CDMA



CDMA

Receiving Data in CDMA



CDMA

Walsh Table

A Walsh table is used to generate orthogonal codes for different stations. These codes allow multiple users to transmit data at the same time without interfering with each other.

A Walsh table is a matrix of codes made of +1 and −1 values. Each row of the table represents a unique code assigned to a user.

The number of sequences in a Walsh table needs to be $N = 2^m$.

$$W_1 = \begin{bmatrix} +1 \end{bmatrix} \qquad W_{2N} = \begin{bmatrix} W_N & W_N \\ W_N & \overline{W_N} \end{bmatrix}$$

a. Two basic rules

$$W_1 = \begin{bmatrix} +1 \end{bmatrix} \qquad W_2 = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix} \qquad W_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$

b. Generation of W_1, W_2, and W_4

Question

Q1. Find the chips for a network with

a. Two stations

b. Four stations

Q2. What is the number of sequences in CDMA if we have 90 stations in our network?

Q3. Assume 4 stations (A, B, C, D) are using CDMA. Each station is assigned a Walsh code of 4 chips. What will be the Combined signal if station wants to transmit following bits

- A sends 1
- B sends 0
- C sends 1
- D sends 0